F1 Podium Prediction — Milestone 1 Report

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Introduction

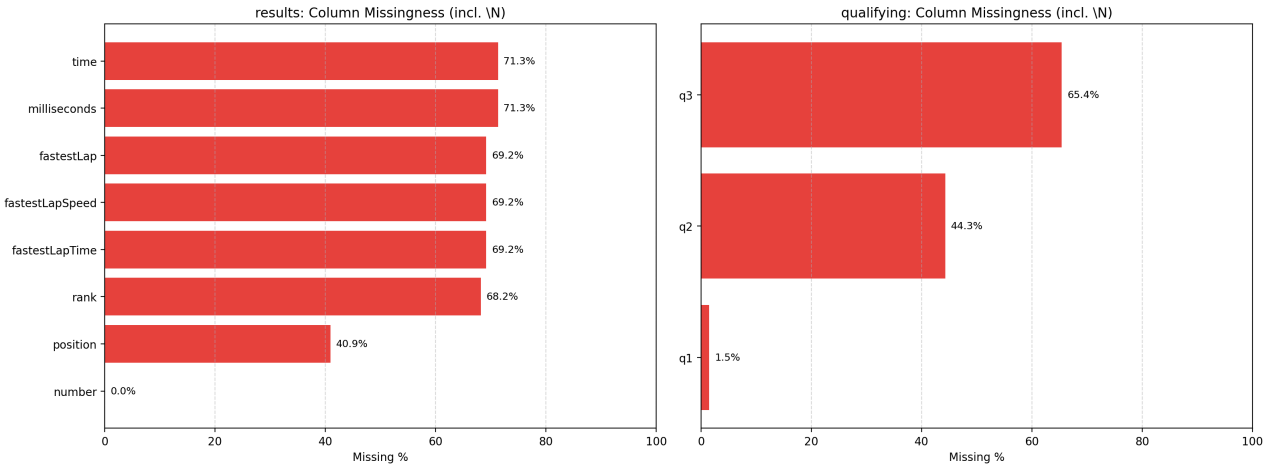
The model predicts, for each driver in each race, whether they finish on the podium (top 3), using only what is known after qualifying and before the start. This report covers Objectives 1–6: cleaning and auditing the data, answering the three data-engineering questions, building the modelling table and its features, training and comparing the models, the feature-group ablation and the explanations of the model. The inference function follows at the end.

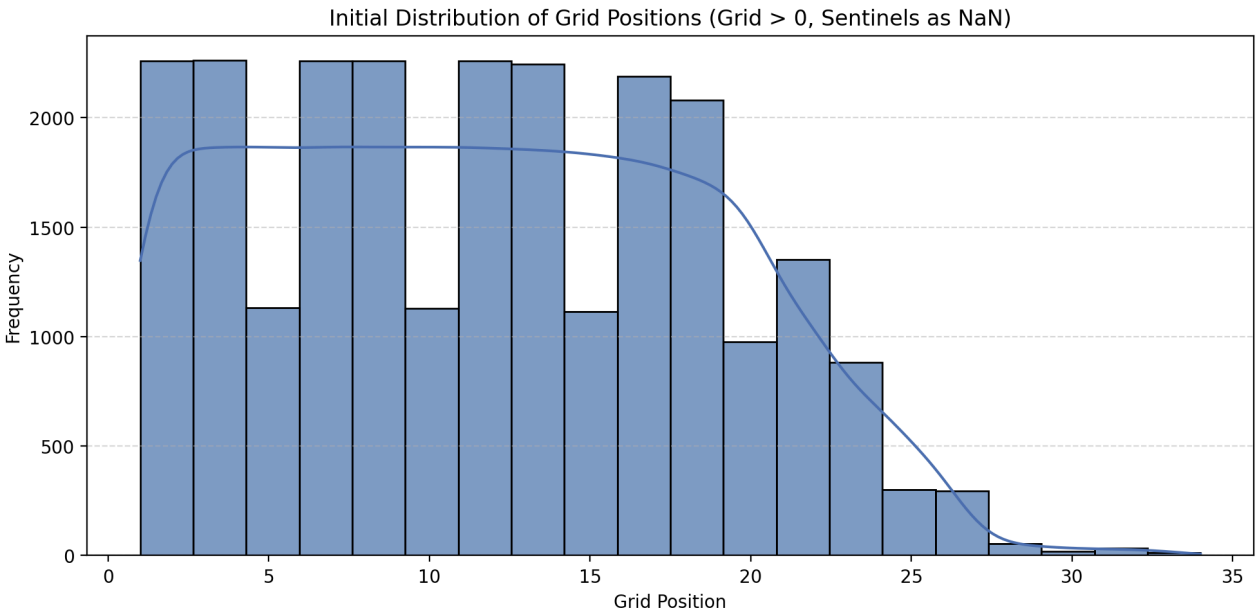
- **Data:** Kaggle "Formula 1 World Championship (1950–2024)" (Ergast), 14 linked tables.
- **Notebook:** one run-all notebook. Sections 1–4 are Objective 1, Section 5 is Objective 2, Section 6 is Objective 3, Section 7 is Objective 4, Section 8 is Objective 5, Section 9 is Objective 6 and Section 10 is the inference function, followed by a short summary.
- **Evidence:** every number and chart in this report comes from that notebook's own run.

1. Data cleaning and auditing (Objective 1)

The raw data held 160,076 missing-value placeholders, wrongly typed columns and 85 duplicated driver–race groups. After cleaning, none remain and every key check passes.

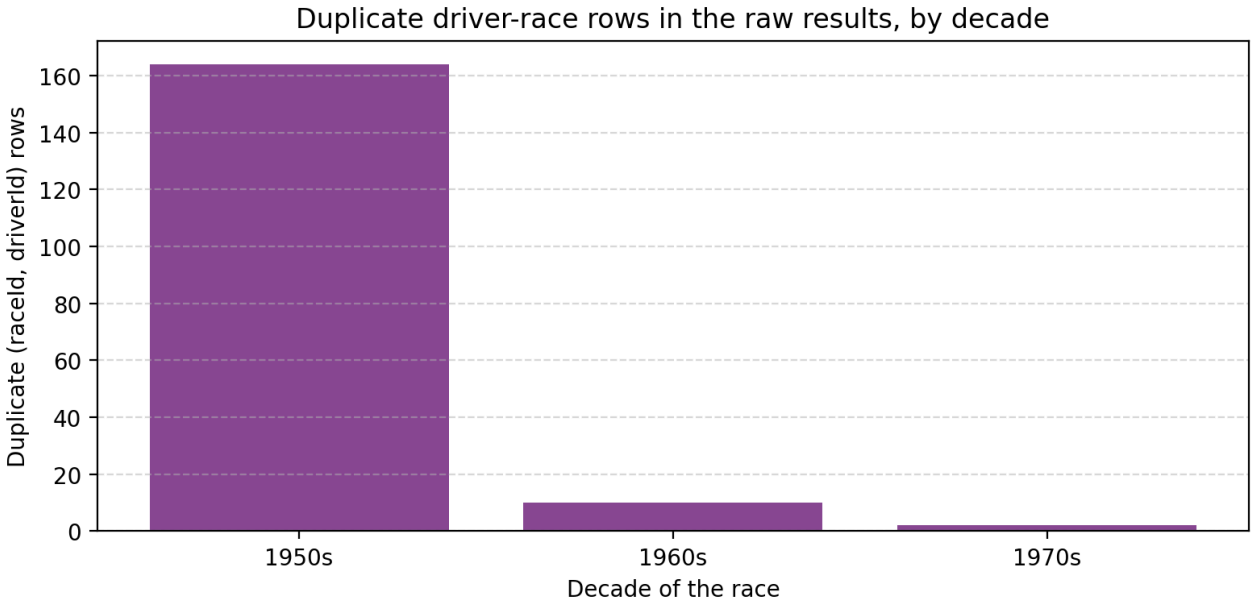
Initial audit. Missing values are stored as the text `\N`, not as empty cells: 160,076 of them, most in `results` (122,887). Timing columns are mostly empty (race time 71.3%, fastest lap 69.2%), and later qualifying sessions are often empty by design (Q3 65.4%). Grid position 0 appears 1,638 times: 92.1% are entries that never started, and no driver has ever reached the podium from grid 0.



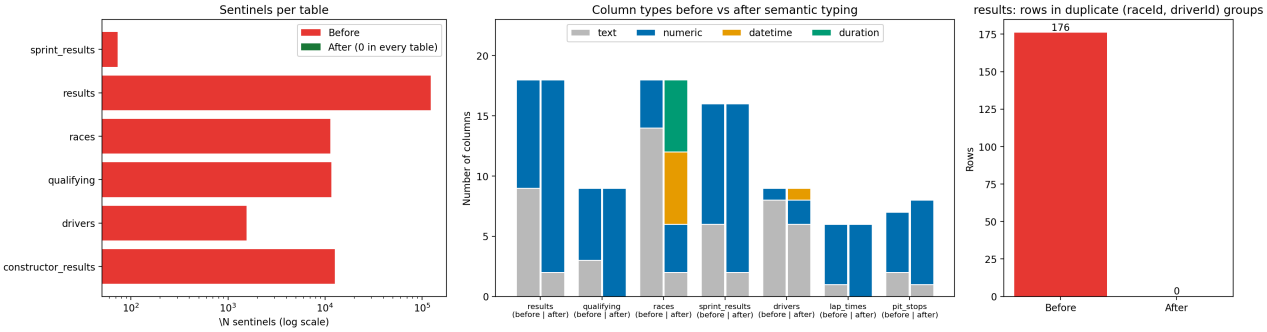


Cleaning. All work is done on copies, so the raw tables stay unchanged.

- **Sentinels:** every `\N` becomes a real missing value.
- **Types:** dates become dates, lap and qualifying times become seconds, and session start times become durations.
- **Duplicates:** 176 rows in 85 (raceId, driverId) groups, all from 1950–1978 (164 of the rows from the 1950s), mostly shared cars. Rule: keep the driver's best-classified row, because a driver credited with a shared podium did score one. Keeping the car the driver started in instead would change the podium label in 17 of the 85 groups.
- **Keys:** all 14 primary keys are unique and non-null, all 23 foreign-key links have 0 orphans, and the per-race tables have no duplicate natural keys.

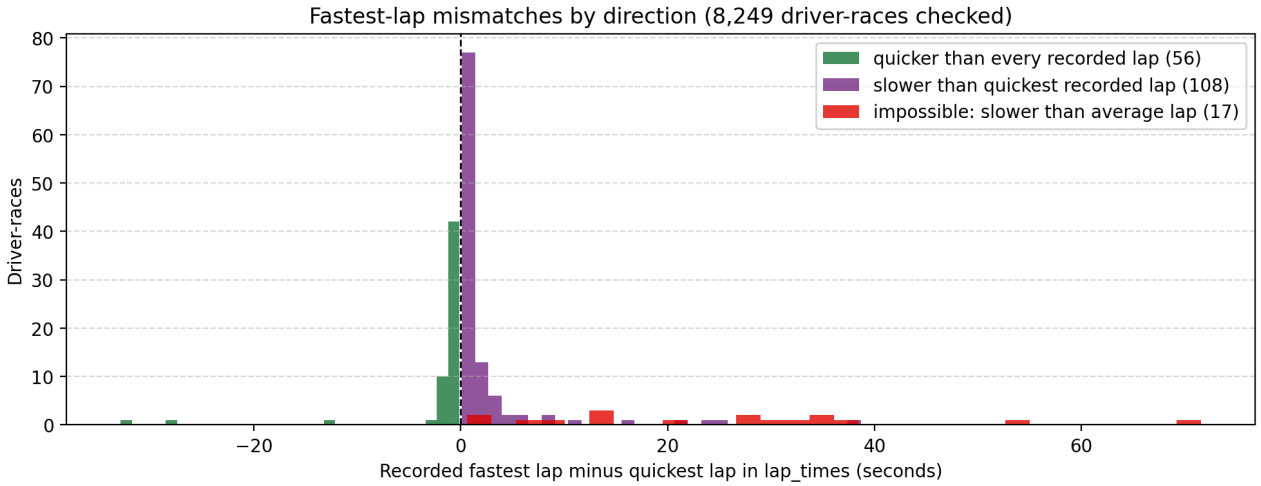


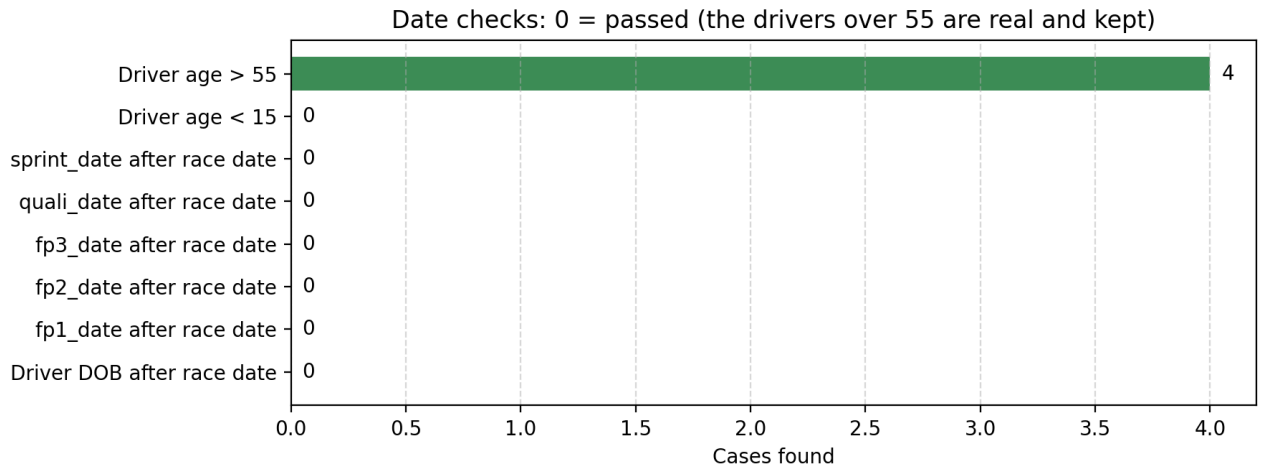
After cleaning. All 160,076 sentinels are gone. Only `results` changes size (26,759 → 26,668 rows, the 91 removed duplicates), and it goes from 9 text columns to 2.



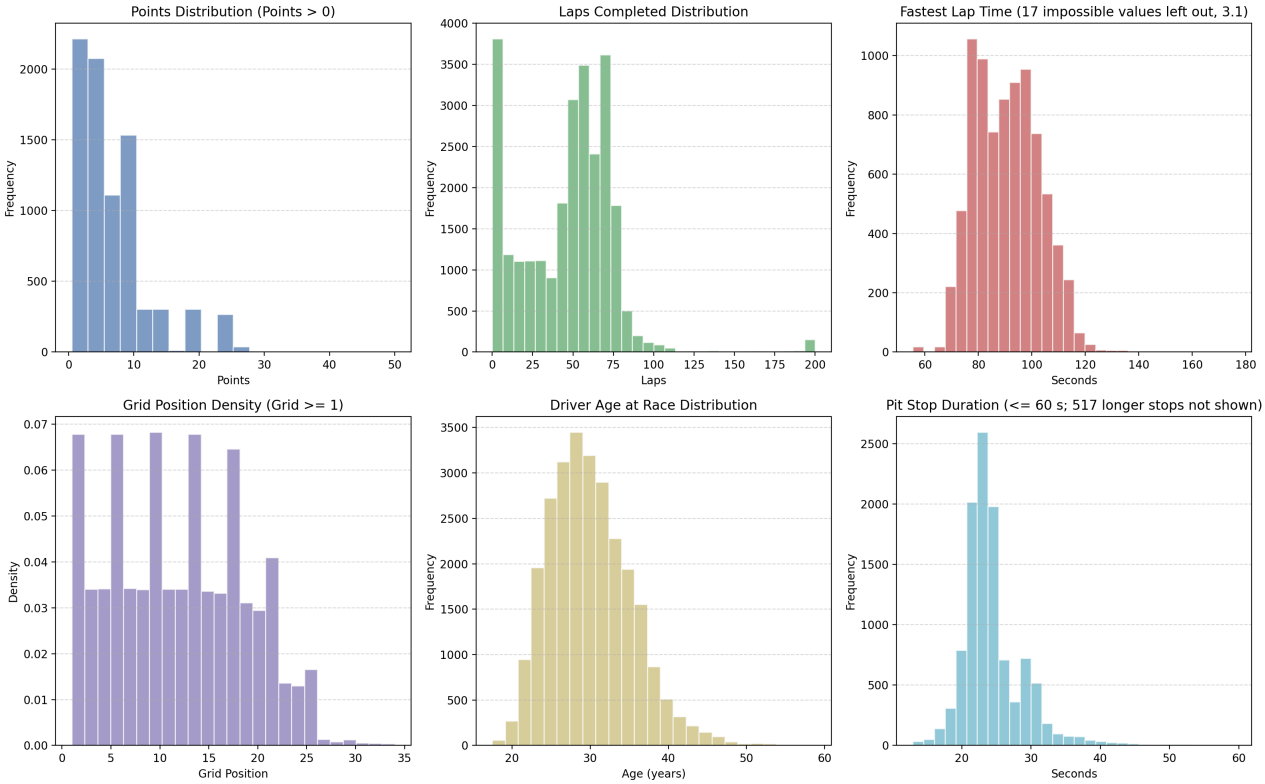
Extra checks.

- **Fastest laps:** of 8,249 driver-races with both records, 17 record a fastest lap slower than their own average lap (impossible), and 181 disagree with the quickest lap in the lap-by-lap data by more than 0.1 s: 56 are quicker than every recorded lap and 108 slower than the quickest one, plus the 17 impossible ones. The data does not show which table is right, so no value is changed: every results row gets a label (`lap_check`), and the fastest-lap chart below leaves out the 17 impossible values. Lap times are race-time information, so they are never model features.
- **Dates:** no date is impossible: no birth after a race, and no practice or qualifying session after its race. Four entries are drivers over 55 (Louis Chiron and Philippe Étancelin in the 1950s); they are real and kept.
- **Before the master merge,** the notebook stops if any `\N` is left, a driver appears twice in a race, or dates and numbers are not typed. All these checks pass.





Distributions. Points follow fixed scales that changed over the decades, so the model uses each driver's share of the leader's points instead. The small peak at 200 laps is the Indianapolis 500.



Master table. One chain of joins starting from `results` gives 26,668 rows, one per driver per race, with no missing keys. One shared rule, `took_the_start`, marks the 1,590 entries that are not race starts: they did not qualify, did not pre-qualify, failed the 107% rule, or withdrew without completing a lap. Cars that broke down on the way to the grid still count as starts, because after qualifying they were expected to race. Every later analysis uses it. Counting starters matters most in the older eras: in the 1980s a race had 28.9 entries but 25.2 starters, so 10.4% of entries but 11.9% of starters reached the podium, against 15.0% for both in the 2020s.

2. Data-engineering questions (Objective 2)

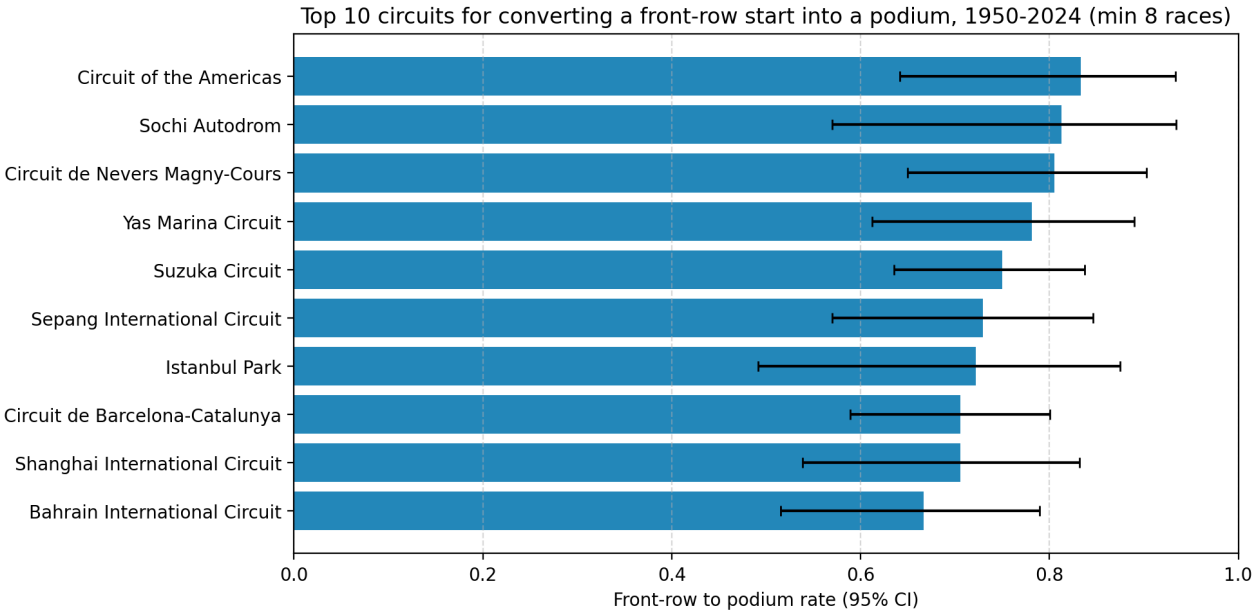
Q1. Which circuits convert a front-row start into a podium most often, and did that change after 2014?

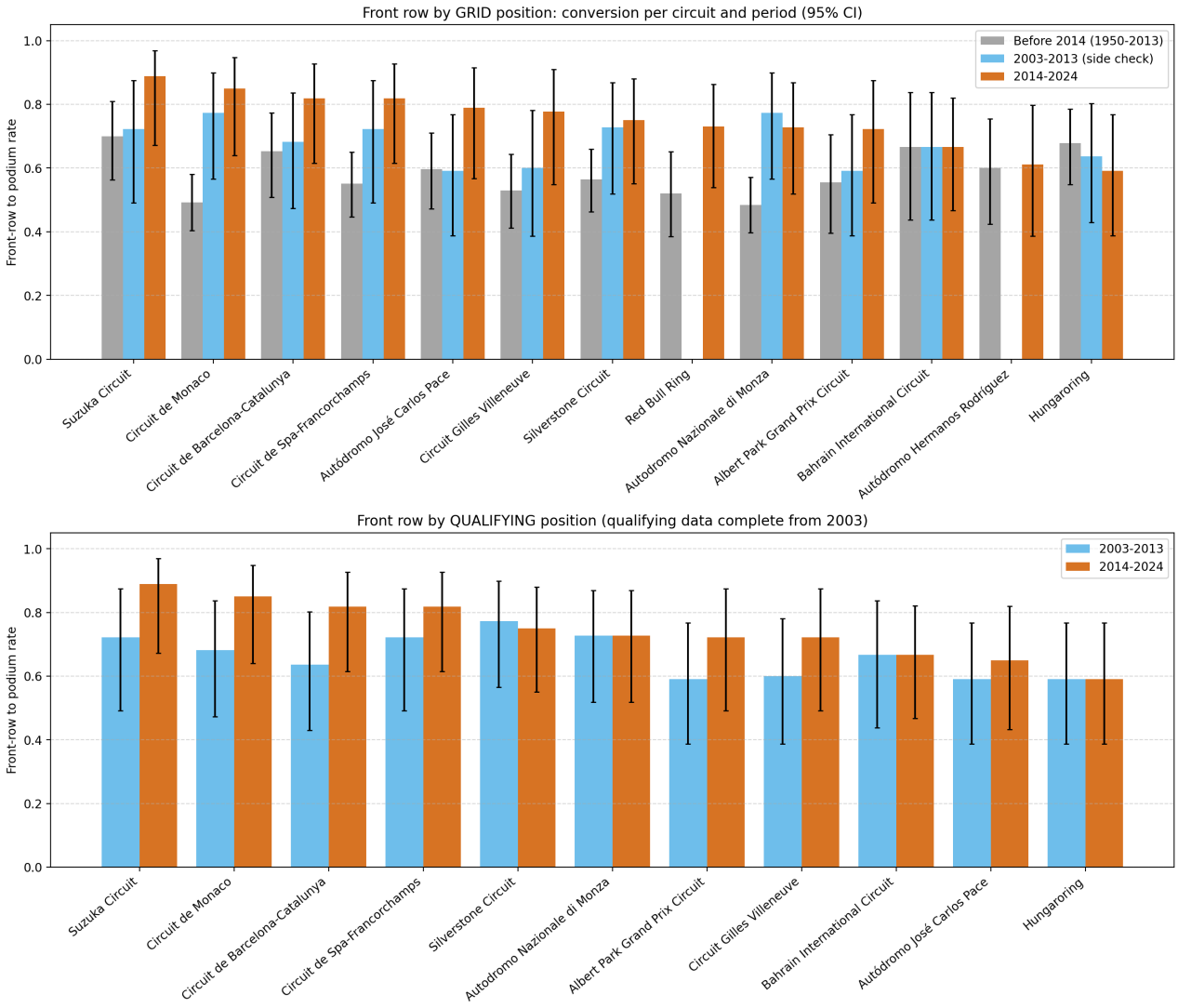
Over 1950–2024 a front-row starter reaches the podium 60.4% of the time. Circuit of the Americas (83%), Sochi (81%), Magny-Cours (81%), Yas Marina (78%) and Suzuka (75%) lead. All ten top circuits first hosted a race in 1987 or later, when front-row cars retired far less often.

Conversion rose from 56.4% (1950–2013) to 76.0% (2014–2024), while front-row retirements fell from 29.8% to 8.4%. The circuit ranking changed a lot (Spearman ρ = −0.13 across 13 circuits). With 2003–2013 as the "before" period, most of that reshuffle disappears (ρ = 0.37 by grid; 0.65 by qualifying, p = 0.03), so much of the change comes from the older decades, not the hybrid era itself.

Grid and qualifying position differ for 5.0% of front-row starters (49 of 984), mainly through penalties and pit-lane starts. Defining the front row by qualifying gives 73.7% for 2014–2024, against 76.0% by grid.

- **Grain:** one row per front-row starter; sample sizes are counted in distinct races.
- **Missing data:** no qualifying positions before 1994 and complete only from 2003, so qualifying-based results use 2003 onwards.
- **Left out:** the Indianapolis 500 (three cars per grid row) and entries that never took the start, 26 front-row entries in total.
- **Caution:** each circuit has only 9–13 races since 2014, so each rate is uncertain by 15–21 points either way.



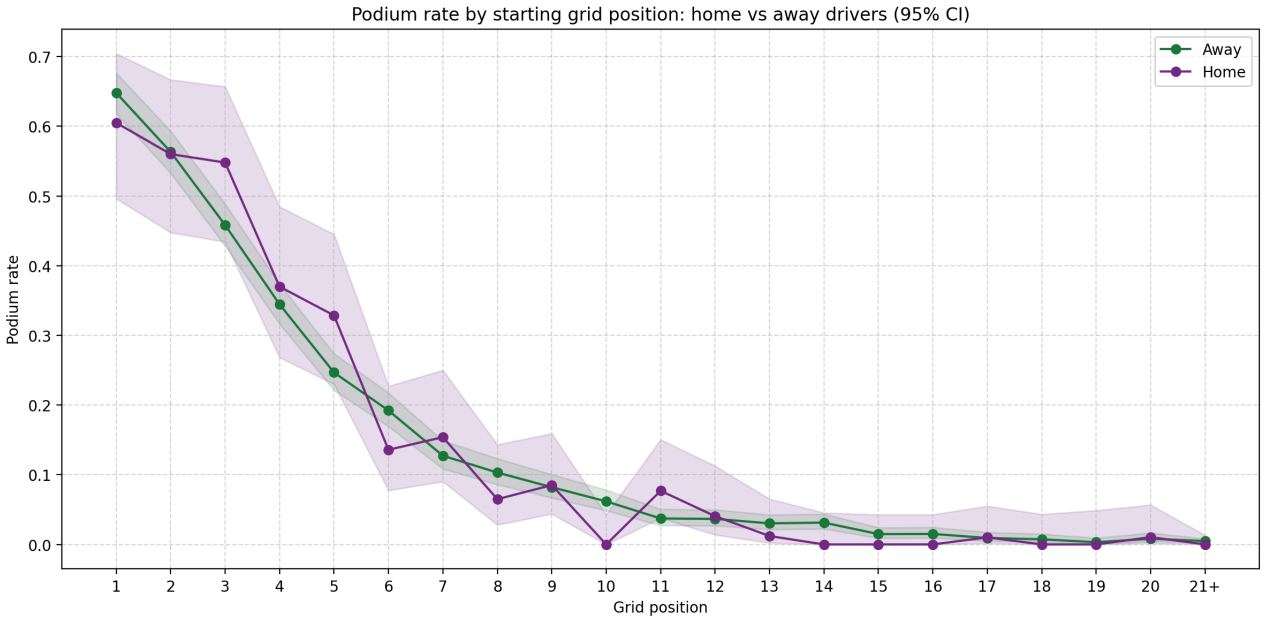


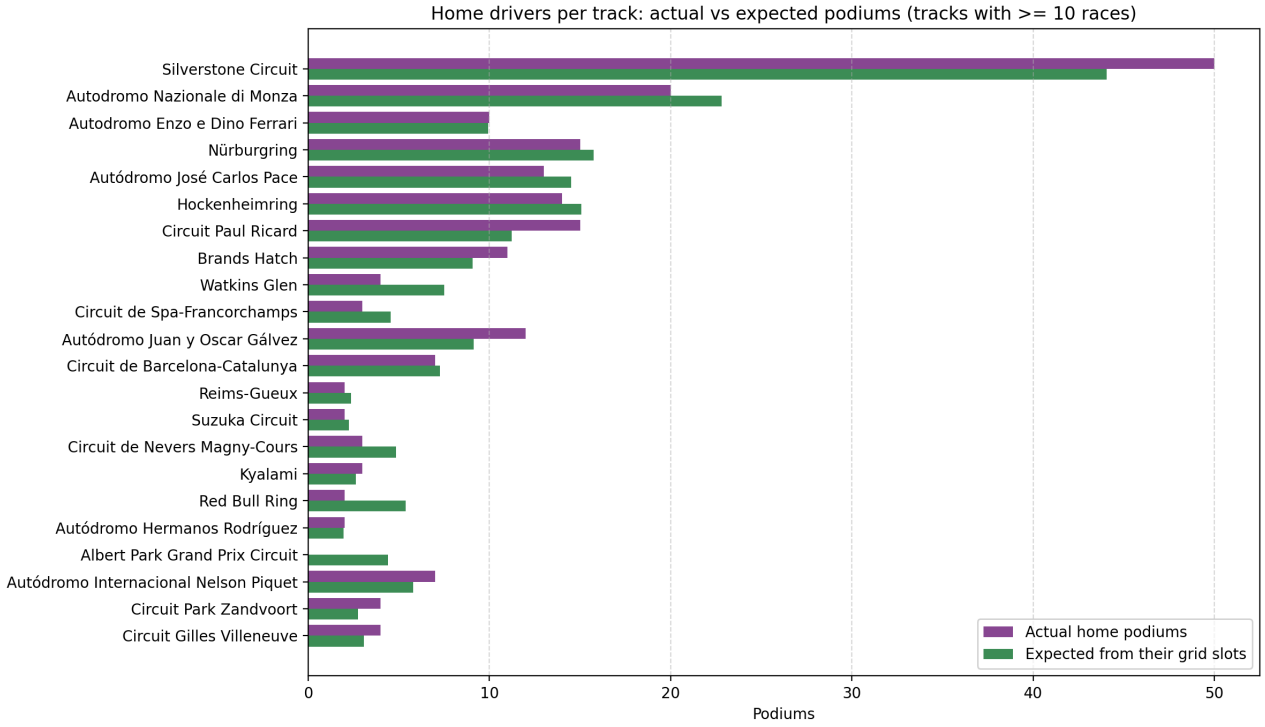
Q2. Do drivers reach the podium more often at home, once starting position is controlled for?

No. Comparing home and away drivers who started from the same grid slot (21 slots combined in a Cochran–Mantel–Haenszel test) gives a pooled odds ratio of 0.97 (95% CI 0.81–1.15, p = 0.71), where 1.0 means no difference.

Raw rates even favour away drivers (13.8% against 11.9% at home), because home entries include many one-off local drivers who start further back. Per track, home drivers scored 230 podiums against 234.7 expected from their grid slots. Only Albert Park stands out (0 against 4.4, p = 0.013), about what chance produces across 22 tracks.

- **Grain:** one row per race start: 24,571 entries with a grid slot, 1,932 of them at home.
- **Missing data:** every nationality maps to a country (0 unmapped), so the home flag is never missing.
- **Left out:** the Indianapolis 500 (381 entries) and 81 entries that held a grid slot but never started.



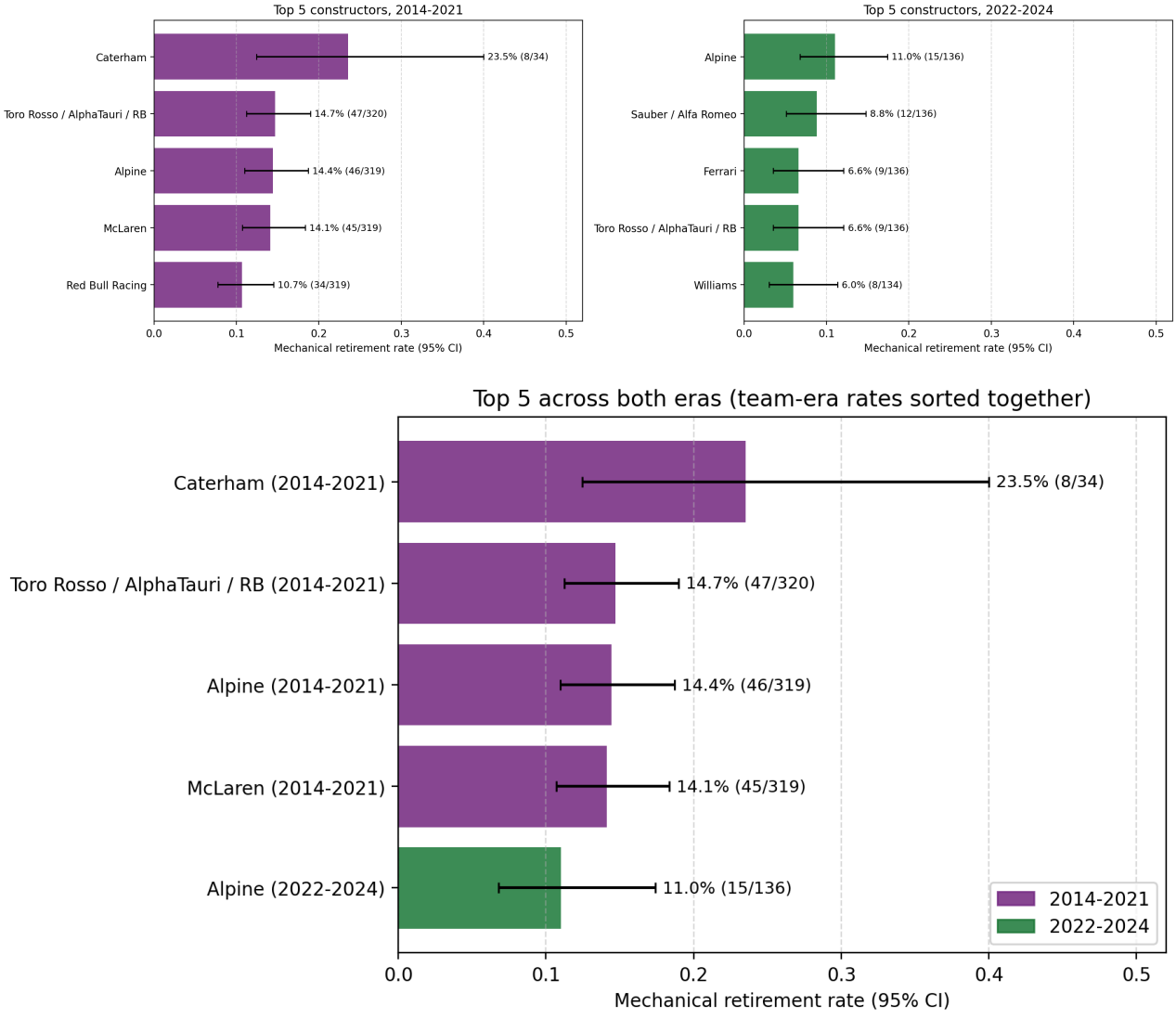


Q3. Which constructors have the highest mechanical-retirement rate in 2014–2021 vs 2022–2024?

In 2014–2021 the highest rates were Caterham (23.5%, from only 34 starts), Toro Rosso / AlphaTauri / RB (14.7%), Alpine (14.4%), McLaren (14.1%) and Red Bull Racing (10.7%). In 2022–2024 they were Alpine (11.0%), Sauber / Alfa Romeo (8.8%), Toro Rosso / AlphaTauri / RB and Ferrari (6.6% each) and Williams (6.0%).

Overall, mechanical retirements fell from 9.39% to 6.13% of starts ($p = 0.00028$). McLaren's drop (14.1% to 3.7%) is the only team change that survives a correction for testing 10 teams.

- **Definition:** every race-ending reason was classified as mechanical, not mechanical or did-not-start; "Wheel" and "Tyre" cases were checked one by one against sources.
- **Grain:** one row per race start; renamed teams are merged.
- **Robustness:** classifying the borderline reasons the other way keeps the same top 5 in both eras.
- **Limitation:** 50 retirements have no recorded reason ("Retired") and cannot be counted as mechanical.



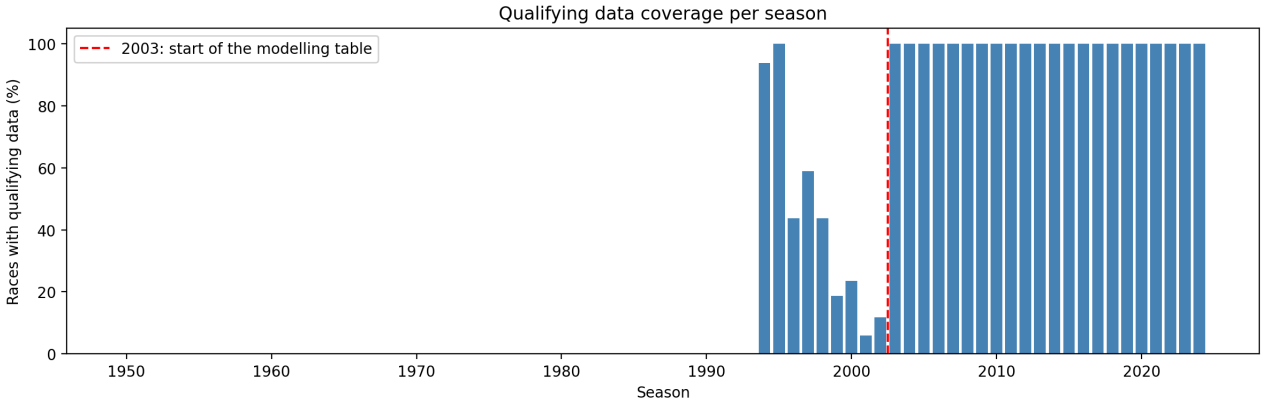
3. Modelling table and data split (Objective 3)

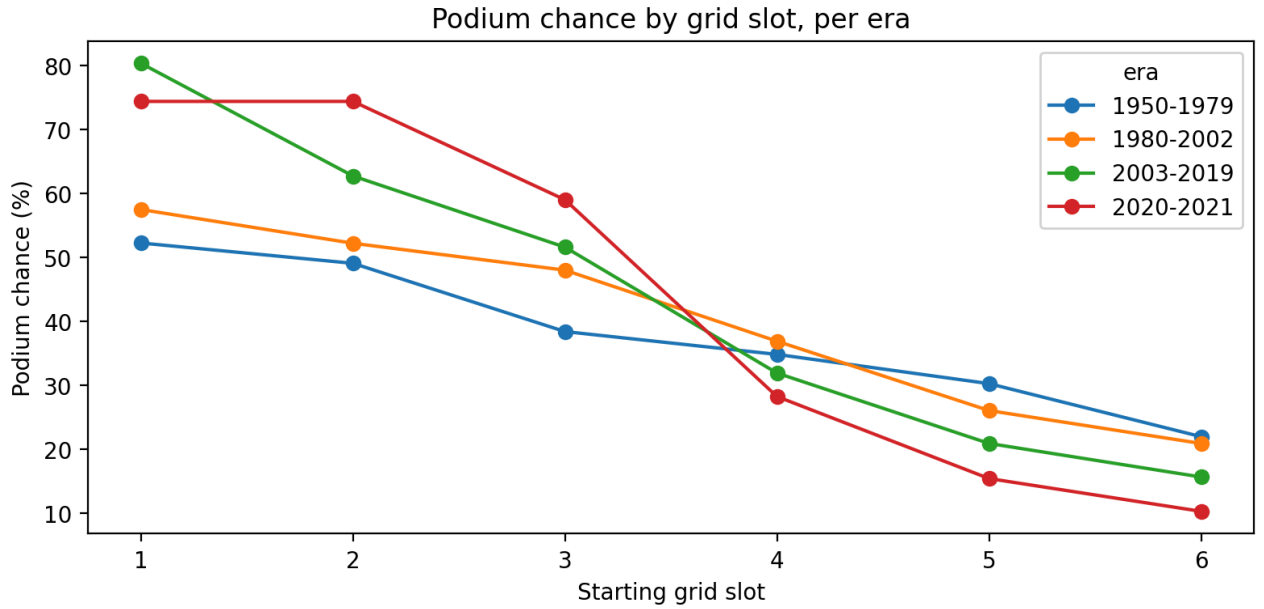
The modelling table has 8,939 rows, one per driver per race start from 2003 to 2024, split by whole seasons into training (2003–2019), validation (2020–2021) and test (2022–2024), the boundaries the course introduction gives.

Rows and target. Only entries that took the start are kept: non-qualifiers and drivers who withdrew before the start can never finish on the podium. Pit-lane starters stay, and the Indianapolis 500 is left

out. The target is podium = finishing position 3 or better, true for 14.4% of rows. Earlier seasons are kept only as history for the form features.

Why 2003. No race has qualifying data before 1994, coverage is patchy until 2002, and every race has it from 2003. Older eras also behaved differently: pole led to a podium 52.2% (1950–1979) and 57.5% (1980–2002) of the time, against 80.3% (2003–2019) and 74.4% (2020–2021, only 39 races). Training on older seasons did not help either (Section 4, experiments).





Split.

Set	Seasons	Rows	Races	Podium rate
Training	2003–2019	6,804	321	14.2%
Validation	2020–2021	780	39	15.0%
Test	2022–2024	1,355	68	15.1%

Whole seasons in time order, for four reasons:

- Drivers in one race compete with each other, so a race is never split across sets.

- A model must not learn from the future.
- Real use predicts upcoming races from past seasons.
- The test seasons are the current car-rules era, so the final score reflects modern F1.

Validation is used for every choice; no choice is made from the test seasons, which give the final scores. The cost of these boundaries is a small validation set: 2020 had only 17 races (a season shortened by COVID) and 2021 had 22, so validation scores are noisy, and Sections 6 and 7 measure how much with 95% intervals.

4. Features, evidence and preprocessing (Objective 3)

The model uses 19 features in 6 groups, all known before the race starts. On its own, the strongest is qualifying position (ROC-AUC 0.910), followed by grid position (0.909).

The features, and why each is valid at the pre-race cutoff. The groups are also the ablation groups for Objective 5.

Group	Feature	Why it is known before the race
Grid	Grid position (a pit-lane start counts as the back of the field)	The starting grid is published before the start
Grid	Pit-lane start flag	Same
Qualifying	Qualifying position	Qualifying happens before the race
Qualifying	Gap to the fastest qualifier, %, capped at 7%	Same
Standings	Driver points share of the leader	From the standings after the previous race, because the standings stored for a race already include it
Standings	Driver championship position	Same
Standings	Team points share	Same; by constructorId, so a renamed team starts again
Standings	Team championship position	Same
Standings	First race of the season flag	Known from the calendar; round-1 standings are last season's
Driver form	Podium rate, last 10 races	Earlier races only; the current race is never included
Driver form	Average finish, last 10 races	Same
Driver form	Retirement rate, last 10 races	Same
Driver form	Rookie flag (first race, no form yet)	Known from the driver's history

Group	Feature	Why it is known before the race
Team form	Podiums per race, last 10 races	Earlier races only; by constructorId, so a renamed team (or an id back after more than a season) starts a new history
Team form	Average best finish, last 10 races	Same
Team form	Retirement rate, last 10 races	Same
Extras	Career starts	Earlier races only
Extras	Past podium rate at this circuit	Earlier visits only
Extras	First visit to this circuit flag	Known from the driver's history

Renamed teams. The course introduction asks that a team feature follow `constructorId` and that a rename resets history, so Toro Rosso and AlphaTauri, for example, count as two teams. The data also reuses a few ids decades apart (Mercedes raced in 1955 and again from 2010), so an id that comes back after more than a season away starts a new history too.

Proof that the features use only the past.

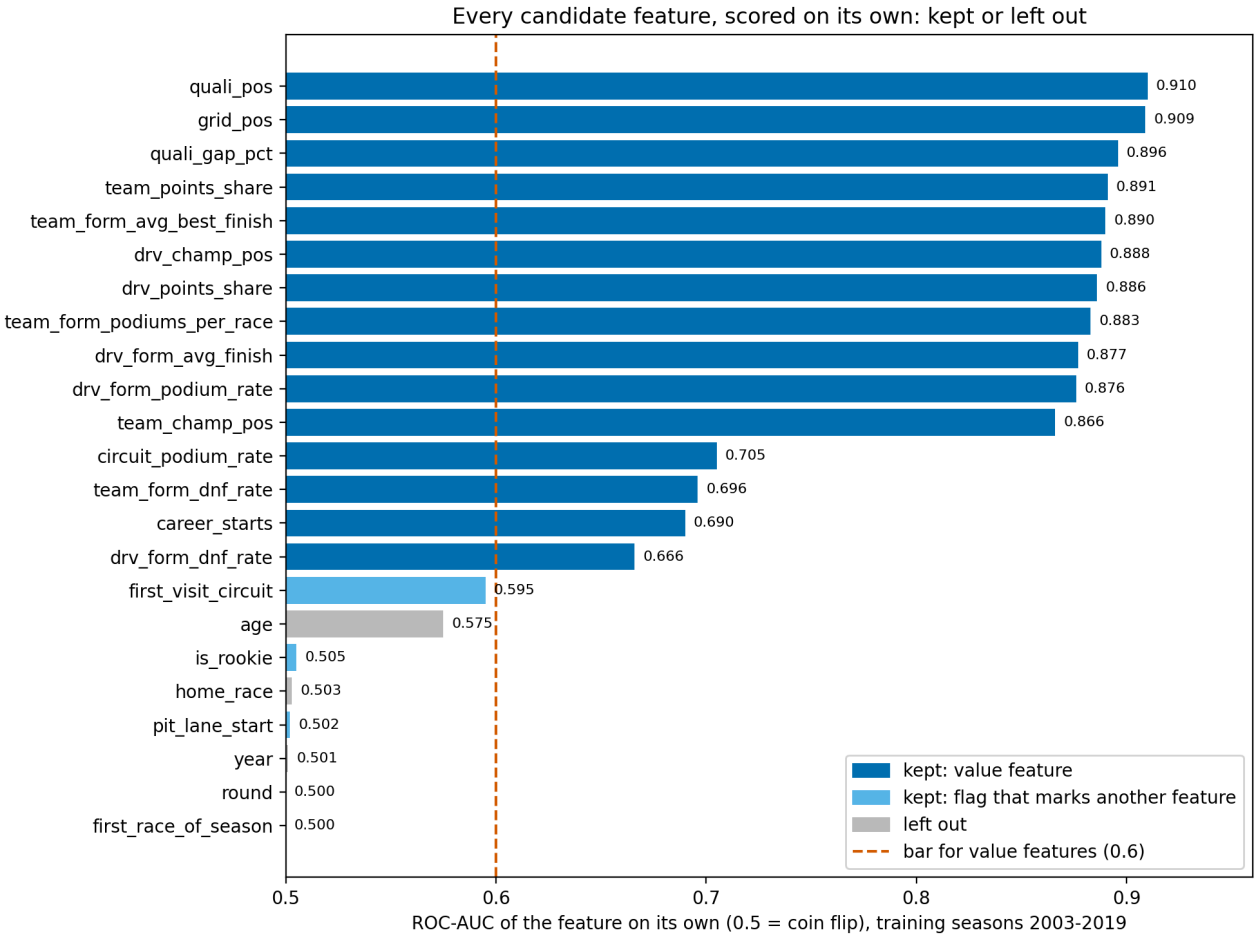
- **Form:** before the 2018 British GP, Hamilton's previous 5 finishes (1, 3, 5, 1, 16) average 5.20, exactly the value in the table; the British GP itself is not used. The example uses the starting 5-race window; the evidence below then picks 10, and the rule is the same for any window.
- **Standings:** the standings stored for that race give the winner, Vettel, 171 points, which already includes the win. The feature uses the 146 he had before it.
- **Checks:** three checks pass on the final table: one row per driver per race, no form value on a driver's first race, and standings always from an earlier race.

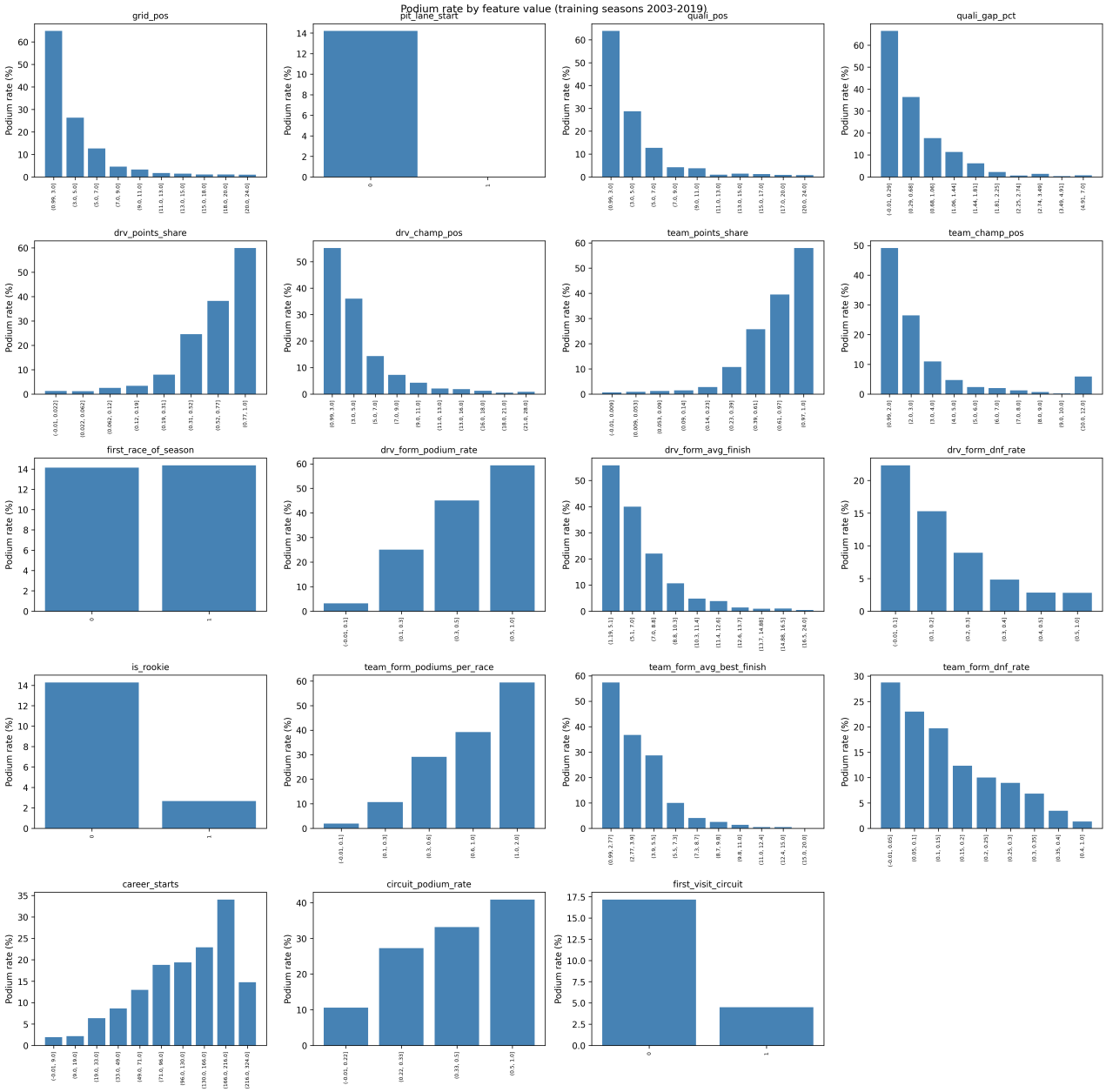
Evidence (training seasons only). Each feature was scored on its own with ROC-AUC: 0.5 is a coin flip, 1 is perfect.

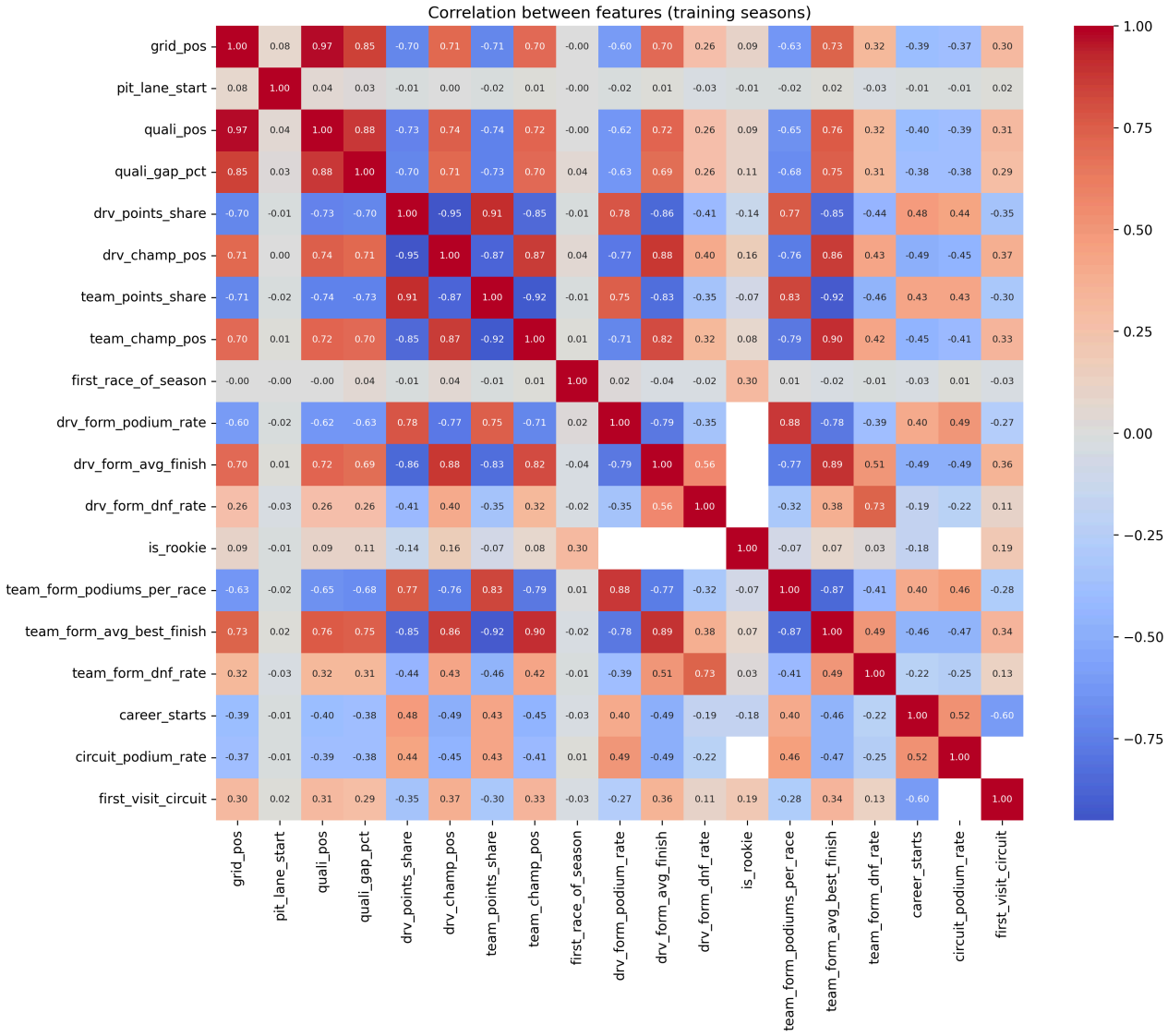
- **Strongest:** qualifying position 0.910, grid position 0.909, qualifying gap 0.896, team points share 0.891, team best finish 0.890, driver championship position 0.888.
- **Useful but weaker:** circuit podium rate 0.705, team retirement rate 0.696, career starts 0.690, driver retirement rate 0.666.
- **Flags near 0.5** (rookie 0.505, pit-lane 0.502, first race 0.500) because they are rare. The charts show that rookies and pit-lane starters rarely or never reach the podium.
- **Left out:** age 0.575, home race 0.503, year 0.501 and round 0.500. Sprint results exist for only 18 races, all in 2021–2024.
- **Form window:** the average score is 0.771 looking back 3 races, 0.796 for 5 and 0.814 for 10, so 10 was chosen.
- **Overlaps:** championship position and points share correlate at −0.95 (driver) and −0.92 (team); grid and qualifying position at 0.97.

Selection rule. One rule decides for every candidate in the scores above. A feature that carries a value (a position, share, rate, count or year) is kept if it scores at least 0.6 on its own, clearly better than a coin flip. A 0/1 flag is kept only if it marks rows where another kept feature was filled in or means

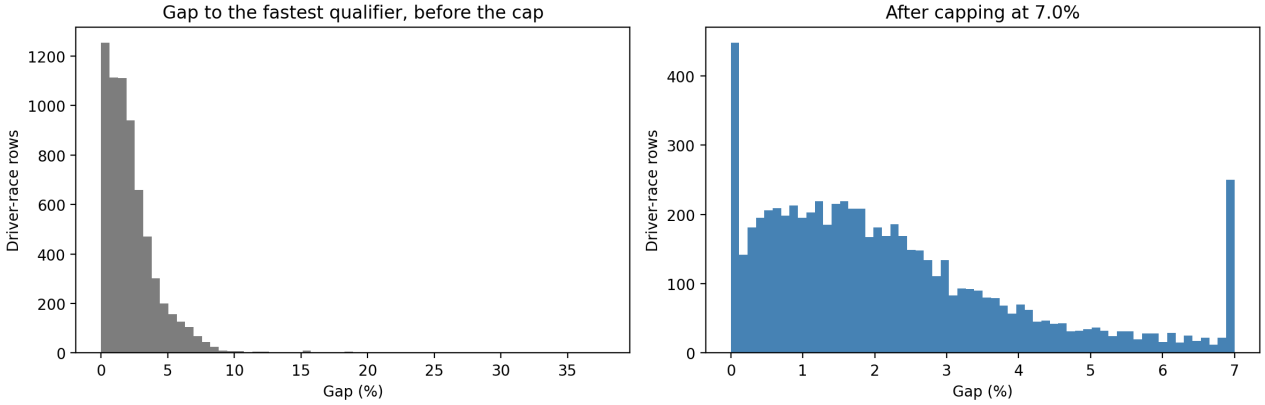
something different: a rookie's form, a first visit's circuit record, a pit-lane grid slot, round-1 standings. The rule keeps exactly the 19 features out of 23 candidates. No value feature scores between 0.575 (age, left out) and 0.666 (driver retirement rate, kept), so any bar in that gap gives the same 19. Home race is a flag that marks nothing else and scores 0.503, so it is left out. Three kinds of candidate are not scored at all: sprint results (no training season has a sprint race), driver or team identity (the model would memorise names) and anything from the race itself (leakage).

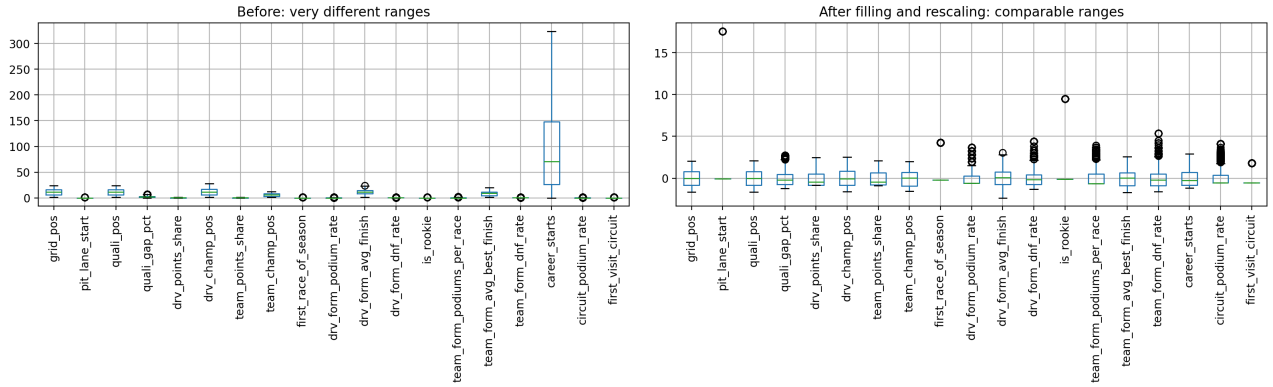



Podium rate by feature value (training seasons 2003-2019)
grid_pos
pit_lane_start
quali_pos
quali_gap_pct
drv_points_share
drv_champ_pos
team_points_share
team_champ_pos
first_race_of_season
drv_form_podium_rate
drv_form_avg_finish
drv_form_dnf_rate
is_rookie
team_form_podiums_per_race
team_form_avg_best_finish
team_form_dnf_rate
career_starts
circuit_podium_rate
first_visit_circuit
Podium rate (%)



Preprocessing. The qualifying gap is capped at 7%, which affects 232 training rows (the largest gap was 37.8%; the capped rows cluster in a few races). Blanks are filled with the training median, and every feature is rescaled to mean 0 and spread 1. The numbers for both steps are learned from the training seasons only, and afterwards no blank values remain in any set. A small reference logistic regression on all 19 features scores ROC-AUC 0.931 and PR-AUC 0.726 on validation.





Experiments. Each changes one thing; the scores are for the validation seasons.

Experiment	Version	ROC-AUC	PR-AUC
Fill and rescale numbers	Learned after splitting (training only)	0.931	0.726
Fill and rescale numbers	Learned before splitting (training + validation)	0.931	0.726
History features	Built on the full timeline	0.931	0.726
History features	Built inside each split	0.930	0.726
Training seasons	From 1950	0.929	0.724
Training seasons	From 1980	0.930	0.725
Training seasons	From 1994	0.930	0.717
Training seasons	From 2003	0.931	0.726
Form window	Last 3 races	0.927	0.724
Form window	Last 5 races	0.930	0.727
Form window	Last 10 races	0.931	0.726

- **Fill and rescale order:** learning the numbers before splitting lets validation shape its own preparation. Here it changes nothing visible, but training-only is the honest order.
- **History features:** building them inside each split makes experienced drivers look new (first visits in validation 28.2% → 65.6%), yet barely changes the score. The full timeline is kept because it describes the drivers correctly.
- **Training seasons:** three times more rows from older seasons does not help; starting in 2003 scores best on both measures, which confirms it.
- **Form window:** the complete pipeline does not clearly prefer a window (differences of 0.004 or less, in opposite directions), so the window chosen in advance on the training seasons (10) stays.

5. Known limitations

The main limits are thinner data in older seasons, changes between car eras, a small validation set, and information the dataset does not hold.

- **Qualifying-coverage drift:** no qualifying data before 1994 and patchy coverage until 2002, so the models learn only from 2003 onwards.
- **Era sensitivity:** the car rules change every few years, and the test seasons (2022–2024) are a new car era. At the start of a season, the standings and much of the form still come from the previous one.
- **Conditions in qualifying:** anything that changes between qualifying sessions, such as rain (not recorded in the data), stretches the qualifying gap; the 7% cap limits the effect.
- **Renamed and returning teams start over:** team form and standings follow `constructorId`, as the course introduction asks, so after a rename, or when an id comes back after more than a season away, a team has no team form for its first race (filled with the median) and no standing at round 1. Team form is blank in 0.7% of training rows.
- **Grid 0 is not always a pit-lane start**: 5 of the 77 grid-0 modelling rows are cars that failed before the start and completed no lap. Right after qualifying they still held their qualifying slot, so for them the grid features use information from after qualifying. None reached the podium, and they are 5 of 8,939 rows, so the effect on the scores is negligible.
- **Small validation set:** 39 races (2020–2021), so differences of a few hundredths between models can be chance; Section 6 measures this with 95% intervals.
- **No model clearly beats the baseline:** the random forest leads the logistic-regression baseline by 0.014 on validation, but the 95% interval (−0.026 to +0.059) includes 0; on the test seasons the two score 0.702 and 0.700 (Section 6).
- **Not used:**
  - driver and team identity, because the model would memorise names, while form and standings already measure strength;
  - sprint results, which exist for only 18 races (2021–2024);
  - anything from the race itself, which would be leakage;
  - weather, tyre choice and practice pace, which are not in the data.

6. Predictive modelling (Objective 4)

Five trained models and a no-model reference were compared on the same rows and the same 19 features. No model clearly beats the logistic-regression baseline on validation, so the baseline is the selected model; on the unseen test seasons it scores PR-AUC 0.700, and the best models 0.702, again within noise.

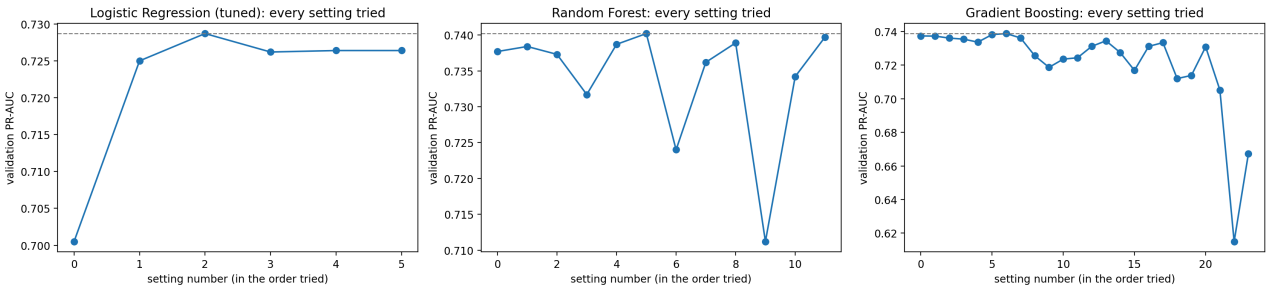
Setup. Every model uses the training, validation and test sets from Section 3, filled and rescaled as in Section 4. Settings, early stopping, the probability threshold and the choice of the model use the training and validation seasons only. The test seasons are scored after every choice is made; Sections 7–9 show test-season numbers only as a check or an example.

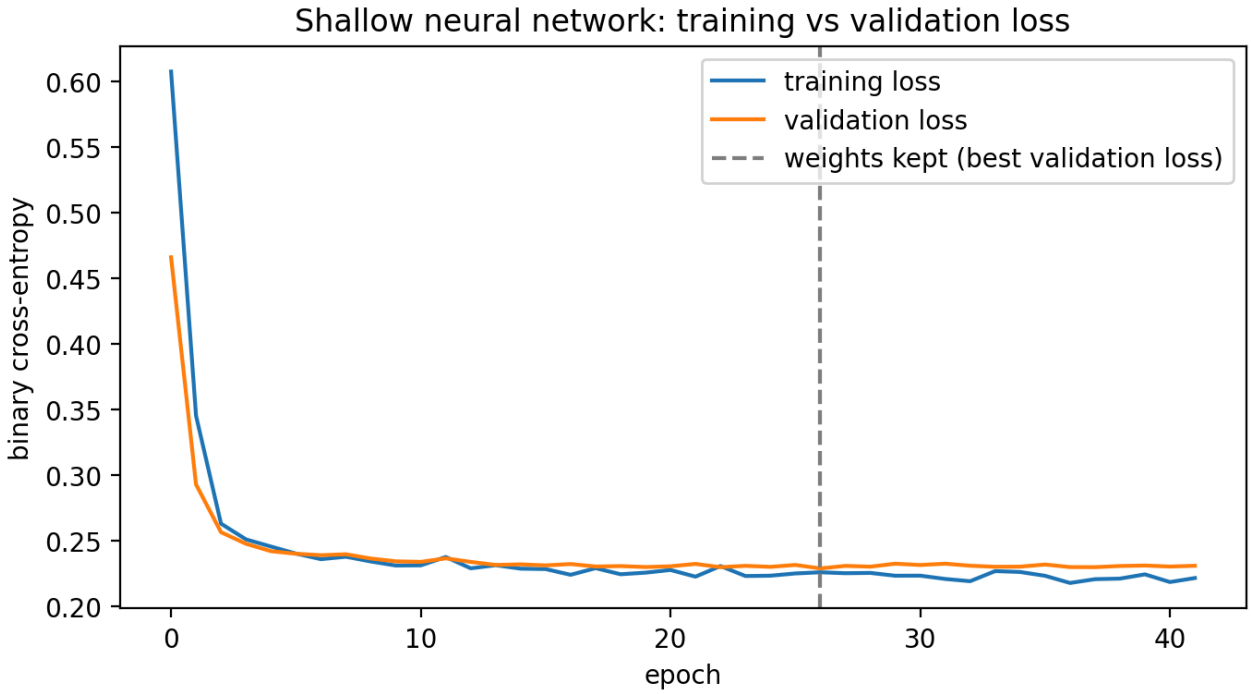
Baseline. A logistic regression with default settings on all 19 features scores validation PR-AUC 0.726 and ROC-AUC 0.931. A no-model reference that ranks drivers by starting position alone scores 0.639 and 0.909. The features' gain over the grid is real: resampling whole races gives the grid rule a gap to the baseline of −0.088 (95% interval −0.136 to −0.043) on validation and −0.053 (−0.098 to −0.004) on test.

Models. Each family was tuned on a small grid of settings, and the setting with the best validation PR-AUC was kept.

Model	Settings tried	Kept (best on validation)	Validation PR-AUC
Logistic regression (tuned)	Penalty strength C: 0.001, 0.01, 0.1, 1, 10, 100 (6)	C = 0.1	0.729
Random forest, 300 trees	Depth 3, 5, 8 or unlimited × at least 5, 20 or 50 rows per leaf (12)	Depth 5, at least 50 rows per leaf	0.740
Gradient boosting	Learning rate 0.03 or 0.1 × depth 2, 3 or 5 × 100 or 300 trees × L2 penalty 0 or 1 (24)	Learning rate 0.03, depth 3, 300 trees, no L2 penalty	0.739
Shallow neural network (PyTorch)	Hidden layers 16, 32, 32 + 16 or 64 + 32 units × dropout 0.1 or 0.3 (8)	Two layers of 32 and 16 units, dropout 0.3, best at epoch 27	0.739

- **Neural network:** ReLU hidden layers, binary cross-entropy loss and the Adam optimiser with a small weight decay. Training stops after 15 epochs without improvement in validation loss, and the best weights are kept. The one-layer networks scored 0.724–0.735, the larger 64 + 32 network 0.713–0.714.
- **No class weights:** the probability threshold handles the 14% / 86% imbalance, and unweighted training keeps the probabilities closer to real podium chances.





How each model works, and its limits here.

Model	How it works	Main limitation here
Grid position only	Ranks drivers by starting position. No learning.	Ignores car and driver strength.
Logistic regression	Weighs each feature, adds the weights up and turns the sum into a probability.	Straight-line effects only; overlapping features (grid and qualifying, 0.97) make single weights unstable.
Random forest	Hundreds of decision trees on random samples of rows and features; their votes are averaged.	Can memorise noise with 321 training races; cannot extrapolate beyond values seen in 2003–2019.
Gradient boosting	Small trees built one after another, each fixing the mistakes of the ones before.	Most sensitive to its settings; same era-shift problem as the forest.
Shallow neural network	Layers of weighted sums with a ReLU bend between them, trained step by step to reduce the error.	Many weights for 6,804 rows, so it needs dropout and early stopping; hardest to explain.

Comparison. Each model gets one probability threshold: the one with the best F1 on the validation seasons, reused unchanged on training and test. The thresholds are 0.17–0.49, mostly well below 0.5, because podiums are rare. The spec's metric table lists ROC-AUC, PR-AUC and F1; precision and recall are added to report five metrics.

Training seasons (2003–2019):

Model	ROC-AUC	PR-AUC	F1	Precision	Recall
Grid position only (reference)	0.909	0.628	0.647	0.649	0.646
Baseline logistic regression (selected)	0.930	0.687	0.630	0.491	0.880
Logistic regression (tuned)	0.930	0.686	0.629	0.489	0.882
Random forest	0.943	0.729	0.669	0.545	0.865
Gradient boosting	0.957	0.778	0.710	0.603	0.863
Shallow neural network	0.938	0.704	0.665	0.673	0.657

Validation seasons (2020–2021, 39 races), where every choice is made:

Model	Threshold	ROC-AUC	PR-AUC	F1	Precision	Recall
Grid position only (reference)	0.333	0.909	0.639	0.692	0.692	0.692
Baseline logistic regression (selected)	0.171	0.931	0.726	0.711	0.605	0.863
Logistic regression (tuned)	0.169	0.931	0.729	0.713	0.604	0.872
Random forest	0.232	0.935	0.740	0.725	0.649	0.821
Gradient boosting	0.268	0.931	0.739	0.711	0.662	0.769
Shallow neural network	0.494	0.927	0.739	0.705	0.727	0.684

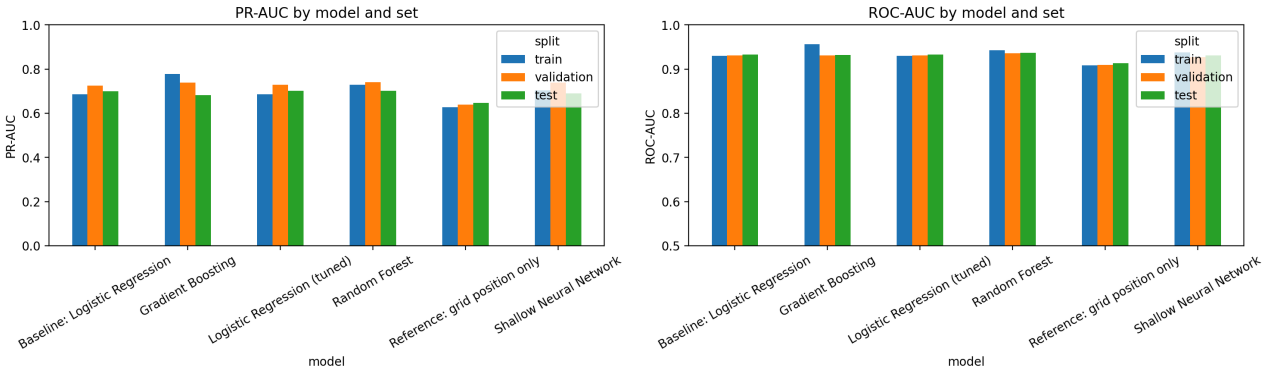
Test seasons (2022–2024, 68 races):

Model	ROC-AUC	PR-AUC	F1	Precision	Recall
Grid position only (reference)	0.914	0.647	0.649	0.650	0.647
Baseline logistic regression (selected)	0.933	0.700	0.633	0.491	0.892
Logistic regression (tuned)	0.933	0.702	0.631	0.488	0.892
Random forest	0.937	0.702	0.655	0.520	0.882
Gradient boosting	0.932	0.683	0.651	0.565	0.770
Shallow neural network	0.931	0.690	0.633	0.692	0.583

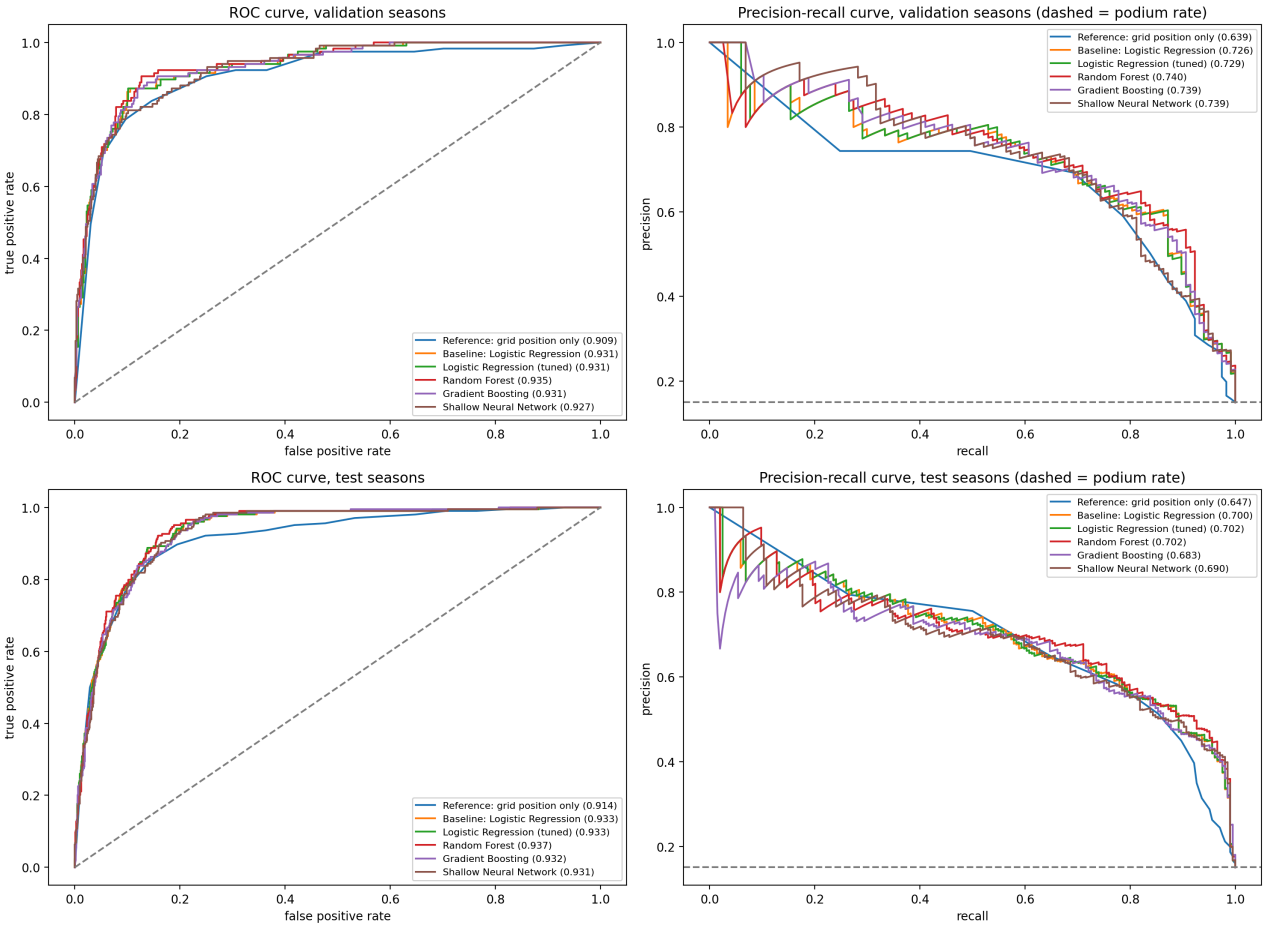
- **Selected model: the baseline.** The rule keeps the simplest model unless another trained model clearly beats it on validation, meaning its whole 95% interval (races resampled 1,000 times) is above 0. The random forest (+0.014, −0.026 to +0.059), gradient boosting (+0.013, −0.027 to +0.056), the network (+0.012, −0.021 to +0.054) and the tuned logistic regression (+0.003, −0.005 to +0.012) do not, so the baseline logistic regression is selected (threshold 0.171).
- **The rule, stated openly:** until the final revision the rule was "highest validation PR-AUC", which would select the random forest. It was replaced on 16 October, after an earlier version of the notebook had already shown test scores, by the rule above, which follows the spec's baseline

requirement. On test the forest and the baseline score 0.702 and 0.700; their gap (+0.002, interval −0.020 to +0.028) includes 0, so the two choices cannot be told apart.

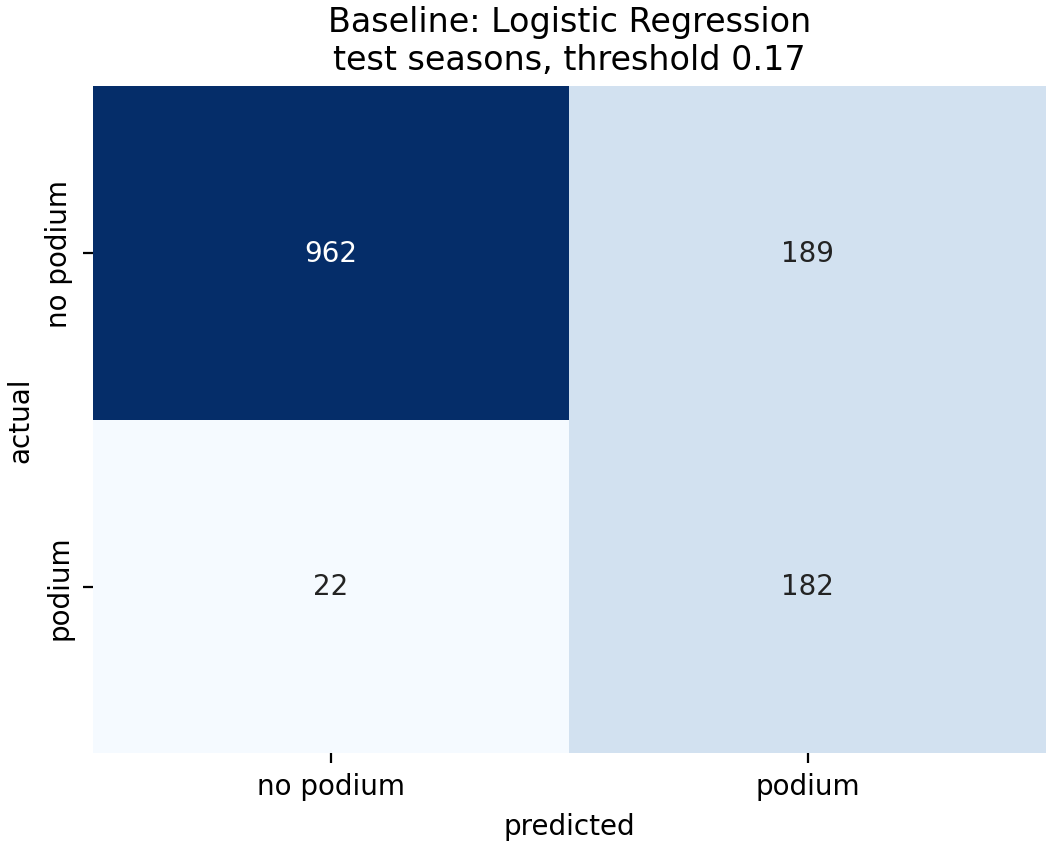
- **Validation to test:** every trained model drops, the flexible ones most: gradient boosting 0.056, the network 0.049, the random forest 0.038, the logistic regressions 0.026–0.027. This fits the era-shift limitation (Section 5). The grid-only rule rises by 0.009.
- **Overfitting:** gradient boosting scores 0.778 on training against 0.739 on validation and 0.683 on test, the clearest overfit. The other models score lower on training than on validation, because the validation seasons happen to be easier.



The ROC and precision-recall curves of the five trained models almost overlap on both sets; only the grid-only rule falls visibly below.



Confusion matrix (test seasons). Of the 204 podium finishers in 2022–2024, the baseline (threshold 0.17) finds 182, misses 22 and wrongly flags 189 drivers. Its low threshold, the F1-best one on validation, trades many false alarms for catching almost every podium finisher.



Primary and secondary metric.

- **Primary: PR-AUC.** Only about 14% of rows are podiums (3 of about 21 drivers per race), so predicting "no podium" for everyone is already 86% accurate. PR-AUC measures how well the real podium finishers are ranked at the top, which is the task. It needs no threshold, so it is fair for choosing settings and models.
- **Secondary: ROC-AUC.** The standard measure of overall separation, comparable with the single-feature scores in Section 4. It is secondary because the many easy non-podium drivers make every model look good: the grid alone scores 0.909.
- **Reported, not used to choose: F1, precision and recall.** They depend on one threshold, itself chosen on validation, and are noisier with 39 validation races. They show what a yes/no decision would look like.

7. Feature-group ablation (Objective 5)

Removing grid and qualifying together is the only change that clearly hurts the model: validation PR-AUC falls from 0.726 to 0.646, and test PR-AUC from 0.700 to 0.583. No single group changes the score by more than chance, because the groups overlap.

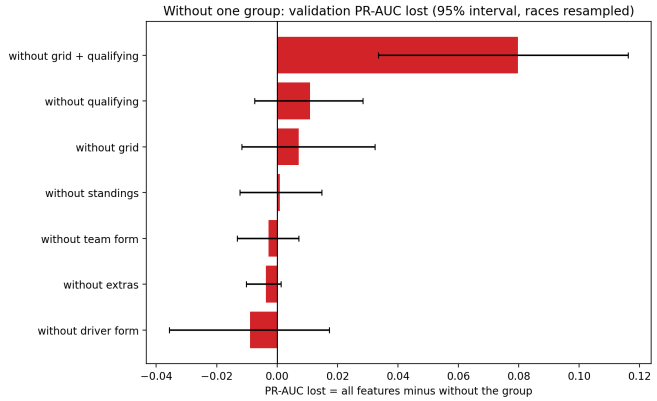
Setup. Each run retrains the selected model (the baseline logistic regression) with the same settings, on the same rows; only the features change. There are two views: without one group, and with one group alone. Grid and qualifying position correlate at 0.97 (Section 4), so one extra run removes both. Each loss gets a 95% interval from resampling the 39 validation races 1,000 times.

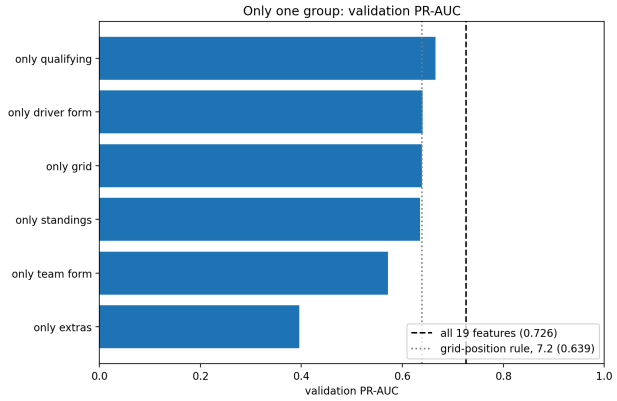
Without one group (validation seasons unless marked):

Run	Features	PR-AUC	PR-AUC lost (95% interval)	ROC-AUC	Test PR-AUC
All 19 features (reference)	19	0.726	0	0.931	0.700
Without grid	17	0.719	0.007 (−0.012 to +0.032)	0.929	0.689
Without qualifying	17	0.715	0.011 (−0.007 to +0.028)	0.928	0.699
Without standings	14	0.725	0.001 (−0.012 to +0.015)	0.932	0.705
Without driver form	15	0.735	−0.009 (−0.036 to +0.017)	0.931	0.700
Without team form	16	0.729	−0.003 (−0.013 to +0.007)	0.931	0.712
Without extras	16	0.730	−0.004 (−0.010 to +0.001)	0.931	0.699
Without grid and qualifying	15	0.646	0.080 (+0.034 to +0.116)	0.903	0.583

One group alone:

Group	Features	Validation PR-AUC	Test PR-AUC
Qualifying	2	0.666	0.647
Driver form	4	0.640	0.562
Grid	2	0.639	0.647
Standings	5	0.635	0.578
Team form	3	0.571	0.502
Extras	3	0.396	0.284





- **Starting position is essential, but either source is enough.** Without both grid and qualifying the loss is 0.080 (test 0.117), the only interval above 0. Without qualifying alone it is 0.011, without grid 0.007: both within noise, because the two carry the same information.
- **Dropping form or the extras even raises the score slightly** (by 0.003 to 0.009), but every interval includes 0: this is noise, not proof that those groups hurt.
- **Alone, most groups are strong:** 0.57–0.67, against about 0.15 for a random ranking; only the extras are weak (0.396). Much of the signal is shared: a fast car shows up in the grid, the standings and the form.
- **ROC-AUC hardly reacts:** removing a single group changes it by at most 0.004, while PR-AUC moves by up to 0.080. This supports PR-AUC as the primary metric (Section 6).
- **Decision:** all 19 features are kept. No single group changes the score by more than chance in either direction, and every group except the extras carries signal on its own (Section 4).

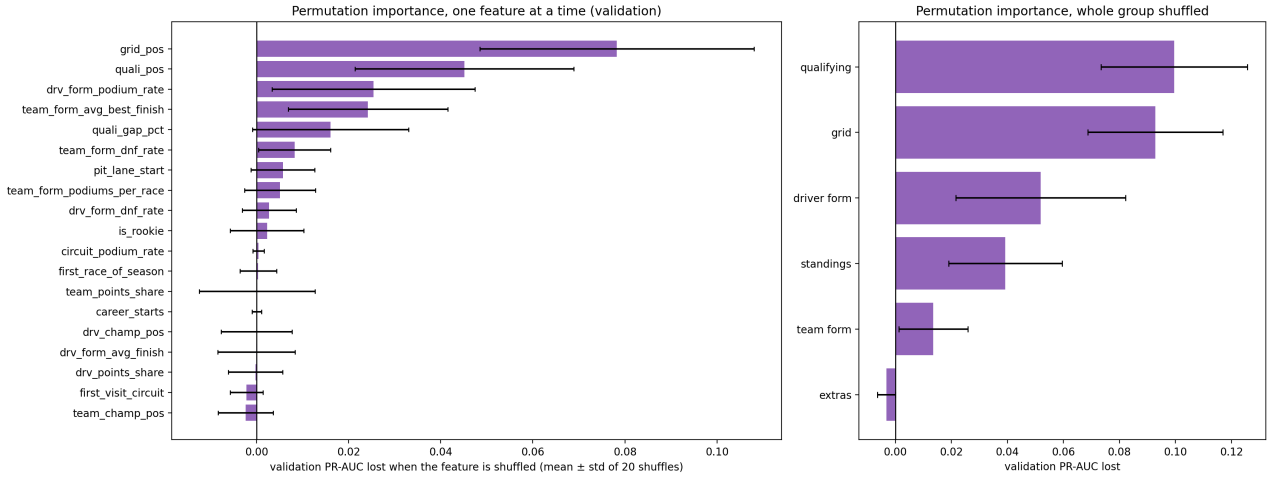
8. Explainability (Objective 6)

The model relies mainly on the starting position (grid and qualifying), then on the driver's recent form and the standings; team form and the extras add little. These results describe what the model uses to predict, not what causes a podium.

Importance is not causation. Every result in this section measures what the selected model relies on. A feature that is important to the model is not proof that it causes podium finishes. A good team best finish matters because a fast car shows up in it; a better finish last week does not make this week's car faster. Qualifying position does not cause a good race result either: both come from the same fast car and driver.

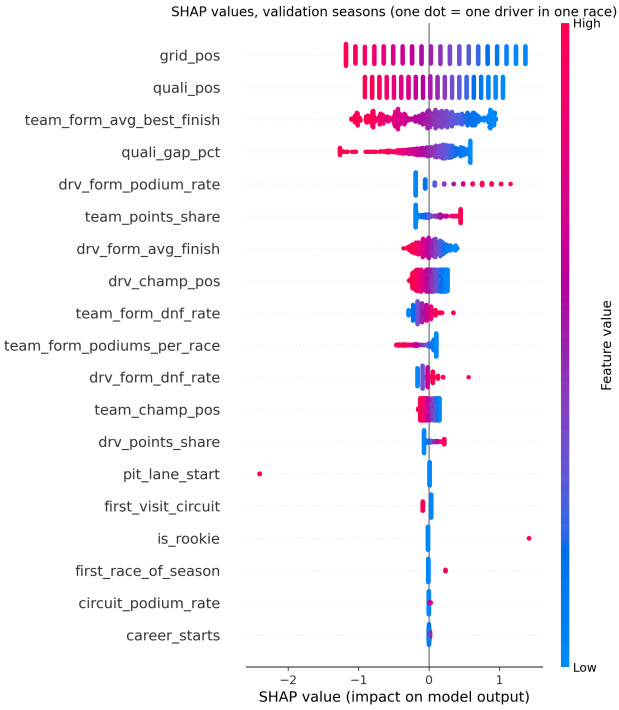
Permutation importance. Each feature's values are shuffled across the validation rows (20 times), so they no longer belong to the right driver, and the fall in validation PR-AUC is measured; the model is not retrained. Whole groups are also shuffled at once, because partners such as grid and qualifying can stand in for each other.

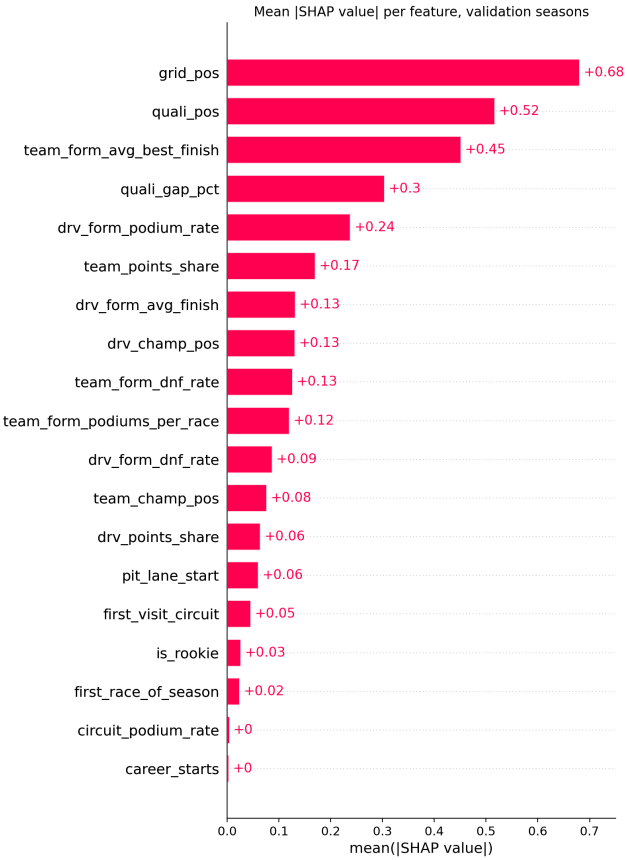
- **Single features:** grid position 0.078 ± 0.030, qualifying position 0.045 ± 0.024, driver podium rate 0.025 ± 0.022, team best finish 0.024 ± 0.017, qualifying gap 0.016 ± 0.017; every other feature 0.008 or less.
- **Whole groups:** qualifying 0.100, grid 0.093, driver form 0.052, standings 0.039, team form 0.014, extras −0.003.
- **Partners hide each other:** no standings feature alone costs more than 0.001, but the whole group costs 0.039, because championship position and points share stand in for each other.
- **Scrambling vs removing:** scrambling qualifying in the trained model costs 0.100, while removing it and retraining costs only 0.011 (Section 7). The model does use qualifying, but a retrained model gets the same information from the grid.



SHAP. SHAP splits every prediction into one contribution per feature, and the contributions add up exactly to the prediction (checked: largest difference 3.6e-15). For a logistic regression the exact method is LinearExplainer: a contribution is the feature's weight × (its value − the training average), in log-odds, the model's score before it becomes a probability. The average output on the training rows is −3.91 log-odds.

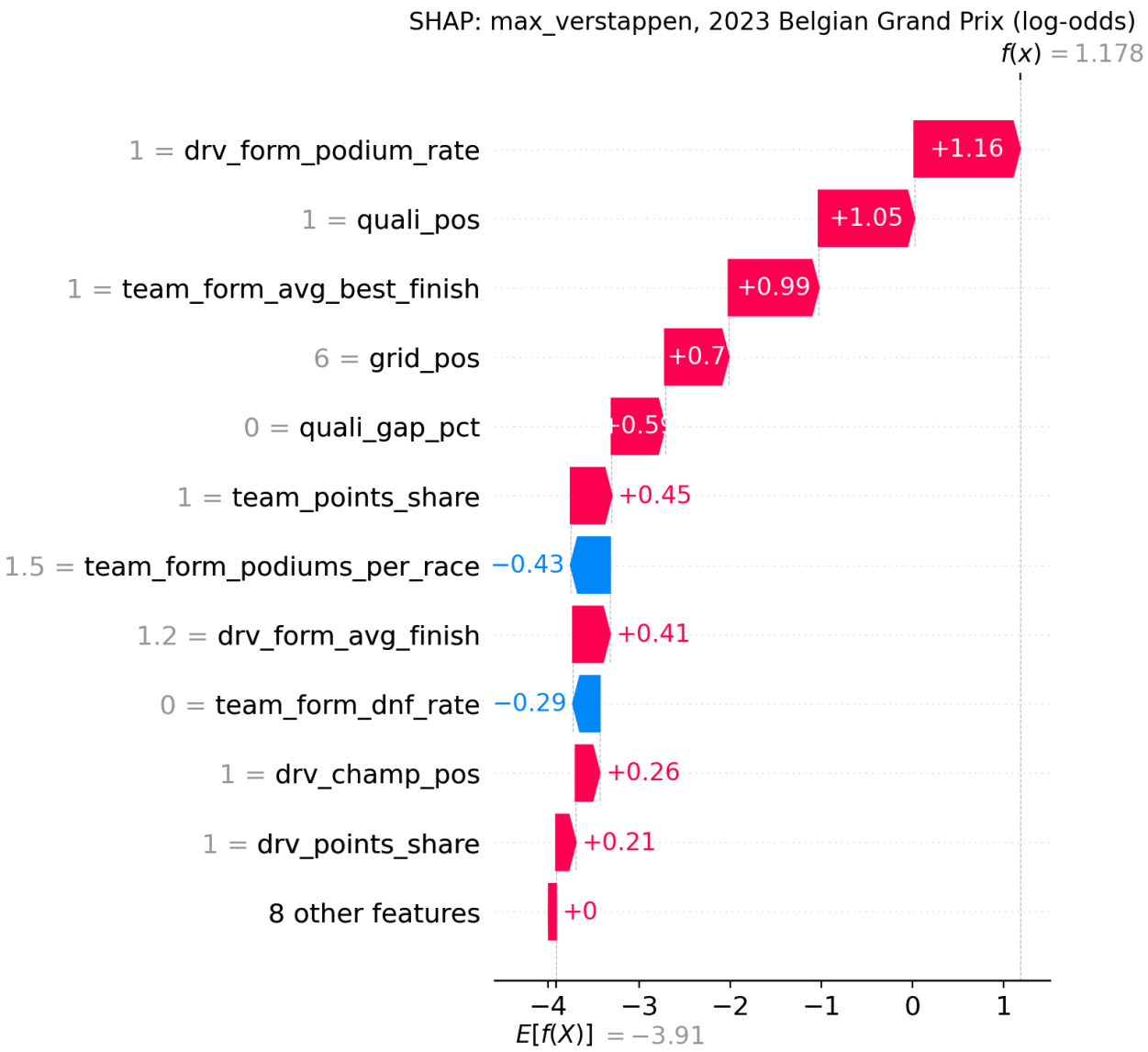
- **Largest average contributions:** grid position 0.680, qualifying position 0.517, team best finish 0.451, qualifying gap 0.304, driver podium rate 0.238, team points share 0.170.
- **Directions match the sport:** a front grid or qualifying slot pushes towards a podium (up to about +1.3), and a good team best finish, a small qualifying gap, a high recent podium rate and a high team points share push up as well; a pit-lane start pushes down by about −2.4.
- **Two surprising signs:** a high team podiums-per-race value pushes down and the rookie flag pushes up. In a linear model overlapping features share and trade credit, so single weights are unstable; the predictions are not affected.

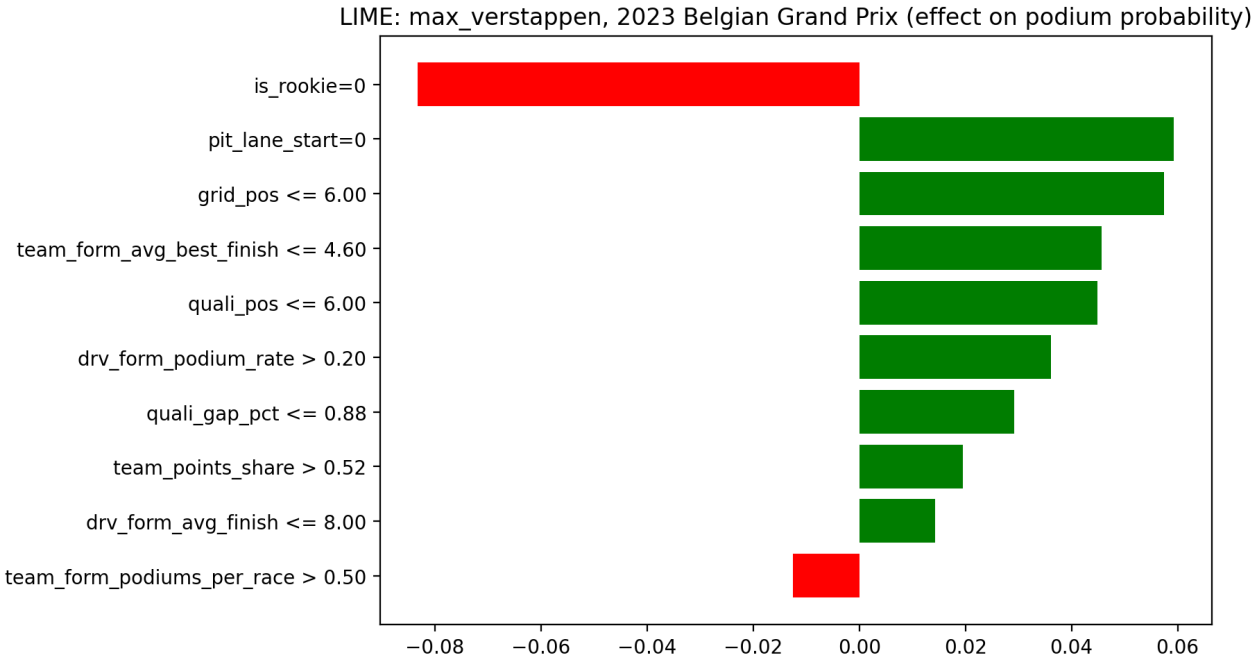




One prediction explained: Max Verstappen, 2023 Belgian Grand Prix. A fixed rule chose it: among test-season drivers who started outside the top 3 and finished on the podium, the one the model was most sure about. He set the fastest qualifying lap but started 6th after a grid penalty, and won. The model gave him a podium chance of 0.76 (threshold 0.17).

- **SHAP:** from the training average (−3.91 log-odds) the contributions add +5.09, reaching 1.18 log-odds, a chance of 0.76. The largest are a podium in each of his last 10 races (+1.16), qualifying position 1 (+1.05) and Red Bull's best car finishing first on average over its last 10 races (+0.99). Grid position 6 adds about +0.7, because 6th is better than the average grid slot.
- **LIME:** 5,000 random rows, each feature drawn from the training data, and a straight-line model fitted to the model's answers, weighted towards the rows most like his. It copies the model poorly here (0.22 against the model's 0.76, R^2 0.47), and its largest weights go to the rookie and pit-lane flags. Its other conditions agree with SHAP: grid position ≤ 6 (+0.057), team best finish ≤ 4.60 (+0.046), qualifying position ≤ 6 (+0.045) and driver podium rate > 0.20 (+0.036).
- **Both point to qualifying pace, recent form and team strength.** SHAP is exact for this model, so its numbers are the ones to quote; LIME's weights are read loosely.


SHAP: max_verstappen, 2023 Belgian Grand Prix (log-odds)
f(x) = 1.178
1 = drv_form_podium_rate
1 = quali_pos
1 = team_form_avg_best_finish
6 = grid_pos
0 = quali_gap_pct
1 = team_points_share
1.5 = team_form_podiums_per_race
1.2 = drv_form_avg_finish
0 = team_form_dnf_rate
1 = drv_champ_pos
1 = drv_points_share
8 other features
+1.16
+1.05
+0.99
+0.7
+0.45
−0.43
+0.41
−0.29
+0.26
+0.21
+0
−4
−3
−2
−1
0
1
E[f(X)] = −3.91



Do the methods agree? Qualifying and grid are the top two in SHAP, in permutation and when a group is dropped; standings and driver form swap third and fourth; team form and the extras are last in SHAP, in permutation and alone (when one group is dropped, every single-group change is within noise). The only real disagreement is which of grid and qualifying carries the starting-position information, and that is interchangeable.

Group	PR-AUC lost when removed	PR-AUC alone	PR-AUC lost when shuffled	Mean SHAP size (log-odds)
Qualifying	0.011	0.666	0.100	0.786
Grid	0.007	0.639	0.093	0.734
Standings	0.001	0.635	0.039	0.421
Driver form	−0.009	0.640	0.052	0.355
Team form	−0.003	0.571	0.014	0.329
Extras	−0.004	0.396	−0.003	0.049

Limits of the explanations.

- **Overlapping features share the credit.** Grid and qualifying position (0.97), championship position and points share (−0.95), and the three team features carry much of the same information; in a linear model the split can even flip a sign, so partners are read together.
- **Small importance means "barely used by this model", not "useless".** Career starts and circuit podium rate barely move any prediction, yet circuit podium rate scores 0.705 on its own (Section 4): its information is already in the other features.

- **The global explanations describe 2020–2021** (39 races). In the new car era of the test seasons the model's reliance may shift (Section 6).
- **LIME is an approximation** (R^2 0.47 here); SHAP for a linear model is exact.

9. Inference function

The inference function scores a race that has not been run yet: for every starter, a podium chance and a yes/no prediction, using only what is known after qualifying. Missing values are filled with the same training medians the models were trained with, so a partially-missing record still gets a prediction, and the function names any feature it was not given.

How it works.

1. The race's starters are taken with their results erased, plus every earlier race as history, and the 19 features are built exactly as in Section 4.
2. Blank values are filled with the training medians and every feature is rescaled with the training numbers.
3. The selected model (the baseline logistic regression) gives the podium chance, and a podium is predicted when the chance is at least 0.171, the threshold chosen on the validation seasons.

For a race that is truly new, its entries (each with its `constructorId`, the team key) and its qualifying, if known, are added first; then the same steps apply.

Demonstration: 2024 Abu Dhabi Grand Prix, the last race in the data, treated as if it had not been run. Rebuilt without its results, all 19 features equal the ones the models were scored on, so the function adds no leakage.

Record	Blank values	Podium chance	Podium predicted	Actual finish
Complete: Norris, on pole	0	0.691	Yes	1st
Partially missing: Doohan, first Formula 1 race (no form, no circuit history)	4	0.005	No	15th
Partially missing: Norris without the two qualifying values	2	0.330	Yes	1st

- **Missing values count as typical.** The blanks get the training medians (podium rate 0, average finish 11.4, retirement rate 0.2, circuit podium rate 0; qualifying position 11, gap 1.809%), so a missing value is treated as neither good nor bad. Without qualifying, Norris's chance falls from 0.691 to 0.330 but stays above the threshold, because pole on the grid and the standings are known.
- **The whole field.** Five drivers pass the threshold (Norris 0.691, Piastri 0.631, Verstappen 0.549, Sainz 0.527, Russell 0.227); the threshold is low and the yes/no rule is per driver, so it can name more than three per race. Two of the five finished on the podium. Leclerc finished 3rd from 19th on the grid with a chance of 0.065, a recovery that a pre-race model cannot foresee.